Laboratory Record
of_____

Roll No.: _____
Experiment No.: _____
Sheet No.: _____
Date: _____

Code:

```
use("college");

db.students.insertMany([

 {

  rollNo: 101,

  name: "Rajesh",

  department: "CSE",

  marks: [

   { subject: "DBMS", score: 85 },

   { subject: "OS", score: 78 },

   { subject: "MongoDB", score: 92 }

  ]

 },

 {

  rollNo: 102,

  name: "Ravi",

  department: "CSE",

  marks: [

   { subject: "DBMS", score: 65 },

   { subject: "OS", score: 70 },

   { subject: "MongoDB", score: 75 }

  ]

 },

 {
```

```
  rollNo: 103,
  name: "Sita",
  department: "CSE",
  marks: [
   { subject: "DBMS", score: 95 },
   { subject: "OS", score: 90 },
   { subject: "MongoDB", score: 98 }
  ]
 }
]);

db.students.aggregate([
 { $unwind: "$marks" },
 {
  $group: {
   _id: "$rollNo",
   name: { $first: "$name" },
   totalMarks: { $sum: "$marks.score" },
   averageMarks: { $avg: "$marks.score" }
  }
 },
 {
  $project: {
   _id: 0,
   rollNo: "$_id",
```



```
    name: 1,
    totalMarks: 1,
    averageMarks: { $round: ["$averageMarks", 2] },
    grade: {
     $switch: {
       branches: [
         { case: { $gte: ["$averageMarks", 90] }, then: "A+" },
         { case: { $gte: ["$averageMarks", 80] }, then: "A" },
         { case: { $gte: ["$averageMarks", 70] }, then: "B" },
         { case: { $gte: ["$averageMarks", 60] }, then: "C" }
       ],
       default: "D"
     }
    }
   }
  },
  { $sort: { rollNo: 1 } }
]);
```

Output:

Laboratory Record

of_____

Roll No.: _____

Experiment No.: _____

Sheet No.: _____

Date: _____

```
{
  name: 'Rajesh',
  totalMarks: NumberInt('510'),
  rollNo: NumberInt('101'),
  averageMarks: NumberInt('85'),
  grade: 'A'
}
```

```
{
  name: 'Ravi',
  totalMarks: NumberInt('420'),
  rollNo: NumberInt('102'),
  averageMarks: NumberInt('70'),
  grade: 'B'
}
```

```
{
  name: 'Sita',
  totalMarks: NumberInt('566'),
  rollNo: NumberInt('103'),
  averageMarks: Double('94.33'),
  grade: 'A+'
}
```

